

# A Generalized Synthetic Control Method for Baseline Estimation in Demand Response Services

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**Abstract**—Accurate baseline estimation is critical for fair settlements in Demand Response and Local Flexibility Markets, yet current regression, clustering, and deep learning methods remain limited in both accuracy and transparency. We introduce a *Generalized Synthetic Control Method* that extends the classical Synthetic Control Method (SCM) from econometrics by incorporating lagged values of both treated and donor units, along with exogenous features, into an augmented control representation. This structure captures temporal dependencies and mitigates correlated residuals that often bias counterfactual estimates. The framework further allows nonlinear weighting through deep neural networks, combining interpretability with expressive power. Benchmarking on the Ausgrid dataset shows accuracy improvements of up to 42%, with linear weighting performing best for large donor pools and nonlinear models enhancing robustness in smaller ones. These results demonstrate that the proposed approach yields more reliable baselines, supporting fair settlements and effective congestion management in flexibility markets.

## I. INTRODUCTION

Flexibility services such as Demand Response (DR) and Local Flexibility Markets (LFM) have become increasingly important for maintaining grid stability amid the greater volatility introduced by Renewable Energy Sources (RES). DR programs incentivize consumers to adapt their energy consumption in response to grid conditions, while LFM provide structured marketplaces where local flexibility is traded to alleviate grid congestion. By leveraging local price signals, they enable cost-effective flexibility procurement, improve resilience, and ensure fair compensation [1].

In both DR and LFM, a central challenge is estimating the *baseline*, the counterfactual consumption that would have occurred absent a flexibility incentive [2], [3]. Delivered flexibility is evaluated as the deviation from the baseline. Hence, accurate and transparent baseline estimation is critical to fair settlements and efficient market functioning. Inaccurate baselines distort payments, discourage participation, or incentivize manipulation through strategic load shifting [4].

Although regression, clustering, and deep learning methods are commonly applied, they often face limitations in accuracy and explainability. To address these challenges, we build on the SCM of Abadie et al. [5], [6] and propose a generalization tailored to baseline estimation. In its classical form, SCM constructs a “synthetic” version of a treated unit

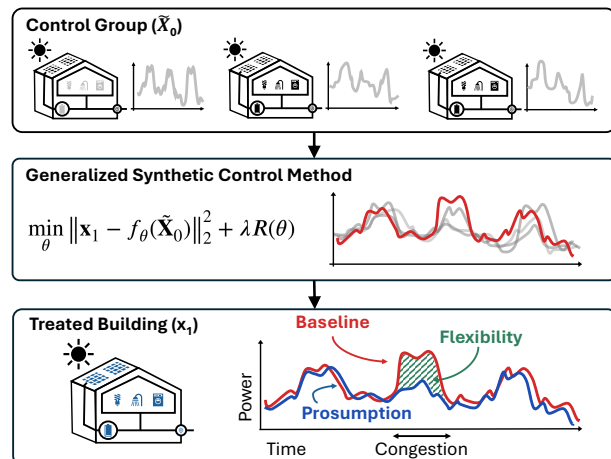


Fig. 1. Architectural overview of the Generalized Synthetic Control Method for baseline estimation in residential flexibility services.

by forming a weighted combination of control units (the donor group) not exposed to the intervention. Applied to our setting, the method estimates a building’s baseline by combining prosumption profiles from similar buildings that did not provide flexibility, with weights optimized to match the building’s historical consumption (Figure 1).

Our contribution extends this framework by incorporating lagged values of both treated and donor units, together with exogenous features, into a constrained nonlinear formulation. This enriched design helps correct correlated residuals and captures temporal dependencies that standard SCM cannot, improving counterfactual accuracy without sacrificing interpretability. The nonlinear extension further broadens the model’s expressive power, yielding stronger performance, especially in smaller or less diverse donor pools.

### A. Related Work

Recent approaches to customer baseline estimation can be broadly categorized into averaging, regression, clustering-based, and deep learning methods. Table I summarizes selected contributions from the literature.

Averaging-based methods, including Exponential Moving Average and X-of-Y techniques, are widely adopted in practice for their simplicity [17]. However, their dependence on historical averages limits accuracy when consumption patterns change rapidly.

Regression-based methods enhance adaptability by modeling consumption as a function of explanatory variables. Early approaches applied Linear and Polynomial Regression

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TABLE I  
CONCEPT MATRIX FOR THE LITERATURE ON BASELINE ESTIMATION IN FLEXIBILITY SERVICES.

| Reference  | Year | Focus                                            | Averaging | Regression | Deep Learning | Clustering | Optimization |
|------------|------|--------------------------------------------------|-----------|------------|---------------|------------|--------------|
| [7]        | 2024 | Aggregated BE with a Graph Neural Network        | ⊙         | ⊙          | ⊙●            | ⊙●         |              |
| [8]        | 2020 | Aggregated BE with Stacked Autoencoders          | ⊙         | ⊙          | ⊙●            | ⊙●         |              |
| [9]        | 2024 | BE with an Attention-based GAN                   | ⊙         | ⊙          | ⊙●            | ⊙●         |              |
| [10]       | 2024 | BE with a Diffusion Model                        | ⊙         | ⊙          | ⊙●            | ⊙●         |              |
| [11]       | 2023 | BE Evaluation of various ML Models               | ⊙         | ⊙          |               |            |              |
| [12]       | 2022 | BE with Mixed Effect Regression                  |           | ⊙●         |               |            |              |
| [13]       | 2022 | BE with K-means and Lasso Regression             | ⊙         | ⊙●         |               | ⊙●         |              |
| [14]       | 2023 | BE with K-means and Weighted Averaging           | ⊙●        |            |               | ●          |              |
| [15]       | 2024 | BE with K-means and LSTM                         |           |            | ●             | ●          |              |
| [16]       | 2020 | BE with Frequency-based Clustering and Averaging | ⊙●        | ●          |               | ⊙●         |              |
| [17]       | 2020 | BE with Dense Neural Networks                    | ⊙         | ⊙          | ●             |            |              |
| This paper | 2025 | BE with Advanced Synthetic Control Method        | ⊙         | ⊙          | ⊙●            | ⊙          | ●            |

Where ● represents a proposed method in the respective category, and ⊙ indicates its inclusion as a benchmark. Abbreviations: Baseline Estimation (BE); Generative Adversarial Network (GAN), Machine Learning (ML), Long Short-Term Memory Model (LSTM).

[17], later extended through regularization techniques such as Ridge and Lasso Regression. More advanced models include Support Vector Machines [11], Random Forests, and XGBoost [17].

Deep learning has substantially advanced baseline estimation by capturing nonlinear dependencies in load profiles. Architectures range from Dense Neural Networks (DNN) and Long Short-Term Memory Models (LSTM) [15] to Autoencoders [8], Generative Adversarial Networks (GAN) with attention mechanisms [9], and Graph Neural Networks (GNN) capturing spatial interdependencies [7]. These methods achieve high accuracy and adaptability but raise concerns about transparency.

Clustering-based methods improve baseline estimation by grouping buildings with similar historical consumption patterns, enhancing accuracy and robustness. Common clustering techniques include K-Means, DBScan, Self-Organizing Maps, and frequency-based clustering using Discrete Fourier Transform [13], [16].

Baseline reporting mechanisms are proposed by [1], [2], [18], designing payment and selection schemes to incentivize participants to truthfully reveal their baseline.

However, the practical implementation of baseline estimation in flexibility programs remains challenging because of inherent requirements for accuracy and interpretability. In this context, the SCM, originally proposed by Abadie et al. [5], [6], offers a structured and transparent alternative. Its causal interpretability and adaptability have led to applications in economics [19], public health, social sciences, and engineering [20].

### B. Contribution and Paper Structure

We propose a Generalized SCM framework for baseline estimation in residential flexibility services. To our knowledge, this is the first work to apply SCM explicitly to baseline estimation. The framework extends the classical SCM by enriching the control representation with exogenous features and lagged consumption values, allowing it to capture temporal structure and eliminate correlated residuals that bias conventional estimates (Appendix B). A nonlinear weighting function further enhances flexibility when relationships

between treated and donor units are complex. Experiments on real-world data show that the proposed method improves baseline accuracy while preserving the interpretability that makes SCM suitable for market applications.

The remainder of the paper is organized as follows: Section II introduces our methodology, while Section III outlines our experimental setup. Building on this, Section IV presents our results, Section V discusses our results, limitations, and future work. Finally, Section VI concludes.

## II. METHODOLOGY

This section presents the methodology, covering control group selection and nonlinear extensions. Throughout, we use the notation ( $y$ ) for scalars, ( $\mathbf{y}$ ) for vectors, and ( $\mathbf{Y}$ ) for matrices.

### A. Baseline Estimation Using the Synthetic Control Method

Building on the Synthetic Control framework, we formulate baseline estimation as a constrained optimization problem that yields interpretable weights and improved accuracy.

Let  $y_{jt} \in \mathbb{R}$  denote the electricity consumption of building  $j \in \{1, \dots, J+1\}$  at time  $t \in T_0 \cup T_1$ , where  $T_0$  and  $T_1$  represent periods without and with grid congestion, respectively. Building  $j=1$  (the treated unit) participates in the flexibility program, while the remaining  $J$  buildings (the donor units) receive no flexibility payments. During congestion ( $t \in T_1$ ), the delivered flexibility  $\hat{\tau}_{1t}$  (the treatment effect) is defined as the deviation of the observed consumption  $y_{1t}$  from the estimated baseline  $\hat{y}_{1t}$  [19]:

$$\hat{\tau}_{1t} = \hat{y}_{1t} - y_{1t}, \quad t \in T_1. \quad (1)$$

The baseline  $\hat{y}_{1t}$  is estimated as a weighted average of the control buildings' consumption:

$$\hat{y}_{1t} = \sum_{j=2}^{J+1} w_j y_{jt}, \quad t \in T_1, \quad (2)$$

where the weight vector  $\mathbf{w} = (w_2, \dots, w_{J+1})^\top$  is chosen to best reproduce the treated unit's pre-congestion profile. Let  $\mathbf{x}_1 \in \mathbb{R}^{|T_0|}$  denote the treated building's consumption in  $T_0$ , and  $\mathbf{X}_0 \in \mathbb{R}^{|T_0| \times J}$  the matrix of donor consumption

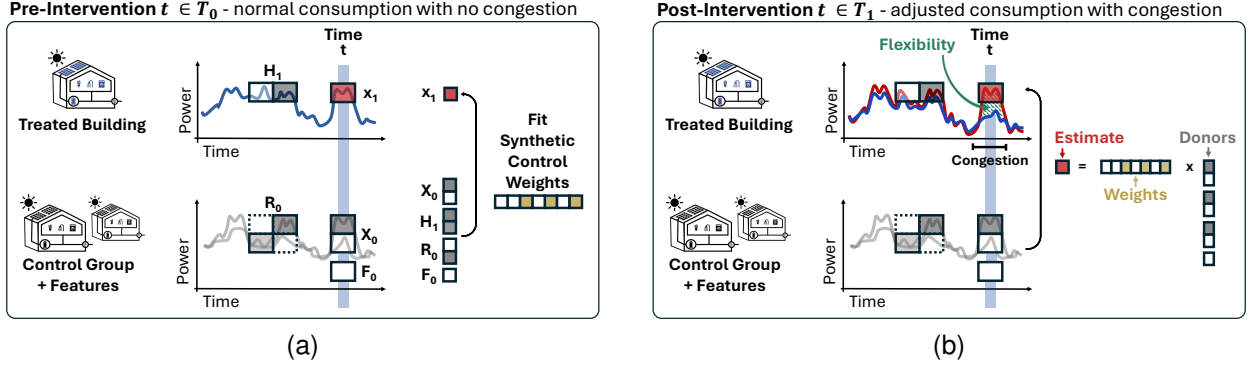


Fig. 2. Overview of the proposed baseline estimation framework for flexibility provision in demand response programs: (a) estimation of the synthetic control weights from the extended control group during the pre-intervention period, and (b) reconstruction of the baseline during congestion events.

over the same period. The optimal weights minimize the reconstruction error:

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \|\mathbf{x}_1 - \mathbf{X}_0 \mathbf{w}\|_2, \quad \text{s.t.} \quad w_j \geq 0, \quad \sum_{j=2}^{J+1} w_j = 1. \quad (3)$$

The non-negativity and simplex constraints ensure interpretability, each  $w_j$  represents the percentage contribution of donor  $j$  to the synthetic control, and act as a form of regularization. In our setting, we relax the non-negativity constraint and allow  $w_j \in \mathbb{R}$  to capture both positive and negative correlations among consumption profiles, while applying explicit regularization to prevent overfitting:

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \|\mathbf{x}_1 - \mathbf{X}_0 \mathbf{w}\|_2^2 + \lambda R(\mathbf{w}), \quad \text{s.t.} \quad \sum_{j=2}^{J+1} w_j = 1. \quad (4)$$

Typical choices for  $R(\mathbf{w})$  include  $\ell_2$ -regularization to promote smooth weight distributions,  $\ell_1$ -regularization to induce sparsity, and hinge-based regularization [21] to encourage balanced donor contributions.

### B. Augmented Synthetic Control Design

Accurate counterfactual estimation in the SCM framework hinges on the donor pool's ability to reproduce both observed predictors and latent factors of the treated unit. In this regard, a key limitation of the standard Synthetic Control approach lies in its static treatment of the pre-intervention matching. In practice, the treated unit rarely lies perfectly within the linear span of the donors; when the resulting errors are temporally correlated, the post-intervention counterfactual becomes suboptimal, as it fails to exploit this predictable temporal dynamic. To address this, we augment the donor group by adding lagged outcomes and contextual covariates, expanding  $\mathbf{X}_0 \in \mathbb{R}^{|T_0| \times J}$  accordingly. See Figure 2 for illustration and also Appendix B for theoretical insights and justification. Specifically, we include:

- (i) **Exogenous features  $\mathbf{F} \in \mathbb{R}^{|T_0| \times M}$ .** For each time  $t \in T_0$ , the corresponding row  $\mathbf{f}_t^\top$  contains  $M$  external covariates (e.g., temperature, humidity, calendar indicators) that serve as proxies for latent demand drivers.

- (ii) **Lagged treated outcomes  $\mathbf{H} \in \mathbb{R}^{|T_0| \times L}$ .** For each time  $t \in T_0$ , the corresponding row  $\mathbf{h}_t^\top$  contains the  $L$  most recent consumption values of the treated unit,  $(x_{1,t-1}, x_{1,t-2}, \dots, x_{1,t-L})$ . Thus,  $\mathbf{H}$  forms a Toeplitz matrix with constant diagonals, as in standard autoregressive (AR/ARX) designs, where  $H_{t,\ell} = x_{1,t-\ell}$  for  $\ell = 1, \dots, L$ .
- (iii) **Selected lagged controls  $\mathbf{R} \in \mathbb{R}^{|T_0| \times J}$ .** For each donor  $j$ , we identify the lag  $k_j^*$  with the strongest absolute Pearson correlation to the treated series, as shown in Equation 5:

$$k_j^* \in \arg \max_{1 \leq k \leq L} |\text{corr}(\{x_{1,t}\}_{t \in T_0}, \{y_{j,t-k}\}_{t \in T_0})|. \quad (5)$$

For each time  $t \in T_0$ , the corresponding row  $\mathbf{r}_t^\top$  contains the selected lagged outcomes from all donors,  $\mathbf{r}_t^\top = (y_{1,t-k_1^*}, \dots, y_{J,t-k_J^*})$ . In this way, each donor contributes its most correlated lagged sequence, capturing both positive and negative dependencies while keeping the feature space tractable for computation.

The augmented control matrix  $\tilde{\mathbf{X}}_0 \in \mathbb{R}^{|T_0| \times D}$  is obtained by concatenating the original donor matrix with the additional feature matrices column-wise,

$$\tilde{\mathbf{X}}_0 = [\mathbf{X}_0 \quad \mathbf{F} \quad \mathbf{H} \quad \mathbf{R}] \in \mathbb{R}^{|T_0| \times D}, \quad (6)$$

where  $D = J + M + L + J$  is the total feature dimension. The synthetic control weights are then estimated using Equation 4, where  $\mathbf{X}_0$  is replaced by  $\tilde{\mathbf{X}}_0$ . Note that standard SCM permits estimation over any post-intervention horizon, as estimators at time  $t$  are independent of treated outcomes after  $T_0$ . When lagged treated values  $\mathbf{H}$  are included, multi-step estimation remains valid only if post-intervention outcomes of  $\mathbf{H}$  are recursively built from estimated counterfactuals rather than realized outcomes. Otherwise, evaluation should be restricted to the first step  $T_0 + 1$ .

### C. Integrating Nonlinear Weighting into the Synthetic Control Method

The classical SCM estimates the counterfactual baseline as a linear combination of control units. This assumption can be

restrictive in practice when the treated unit does not lie (even approximately) in the linear span of the donors. We therefore replace the linear weighting with a function  $f_\theta : \mathbb{R}^D \rightarrow \mathbb{R}$  that maps the  $i$ -th row of the augmented control matrix  $\tilde{\mathbf{X}}_0$ , denoted  $\tilde{\mathbf{x}}_{0i}$ , to the treated outcome  $x_{1i}$  at each time  $t$ .

We estimate  $\theta$  on the pre-treatment window  $T_0$  by minimizing the regularized mean squared error:

$$\theta^* \in \arg \min_{\theta} \frac{1}{|T_0|} \sum_{i \in T_0} (x_{1i} - f_\theta(\tilde{\mathbf{x}}_{0i}))^2 + \lambda R(\theta), \quad (7)$$

where  $R(\theta)$  encodes model regularization.

During congestion periods  $t \in T_1$ , the counterfactual baseline is obtained by a forward pass of the trained model  $f_{\theta^*}$  with the augmented control features as the input:

$$\hat{y}_{1t} = f_{\theta^*}(\tilde{\mathbf{x}}_{0t}), \quad t \in T_1. \quad (8)$$

So, Equations (7) and (8) generalize SCM while allowing nonlinear dependencies and improved fit under complex demand patterns. In practice,  $f_\theta$  can be implemented using complex neural network architectures tailored for sequence modeling (e.g., LSTM or Transformer) to better capture the temporal characteristics of electricity consumption data. In this paper, however, our goal is to introduce and emphasize the general framework rather than optimize the network architecture, and therefore we only use basic Multilayer Perceptron (MLP) networks. We discuss in Appendix B how the augmented SCM already eliminates correlated errors without neural networks. Note also that the classical linear SCM is recovered by the special case  $f_\theta(\mathbf{z}) = \mathbf{z}\theta$  (with  $\mathbf{z} = \tilde{\mathbf{x}}_{0i}$ ).

### III. EXPERIMENTAL SETUP

#### A. Datasets

We use the Ausgrid dataset [22], containing half-hourly measurements of electrical load and Photovoltaic (PV) generation (kW) from 300 Australian households between 2010 and 2013. To extend this dataset, we simulate Battery Energy Storage Systems (BESS) operation via a linear programming model (Appendix VI-A), which optimizes charging and discharging schedules under dynamic electricity prices. This yields complete presumption time series combining household load, PV generation, and BESS behavior. Meteorological variables from Meteostat [23] are spatially and temporally aligned with the Ausgrid data to capture exogenous effects. The final dataset is split into training (60%), validation (10%), and testing (30%) sets and is publicly available at [24].

#### B. Model Configurations and Hyperparameters

For the linear SCM, we evaluate combinations of the  $\ell_2$  norm with  $\ell_2$ ,  $\ell_1$ , and hinge-based regularization. The best results are obtained with  $\ell_2$  regularization, using  $\lambda = 450$  for the augmented and  $\lambda = 1350$  for the basic control group to promote uniform weights. Comparable performance is achieved with hinge-based regularization ( $\lambda = 300$ ,  $\delta = \frac{1}{j}$ ). Unless otherwise noted, the basic control group includes all  $J = 299$  donor units. The augmented control group further adds  $L = 336$  recent treated-unit values, one selected lagged sequence per donor, and  $M = 7$  exogenous features: four

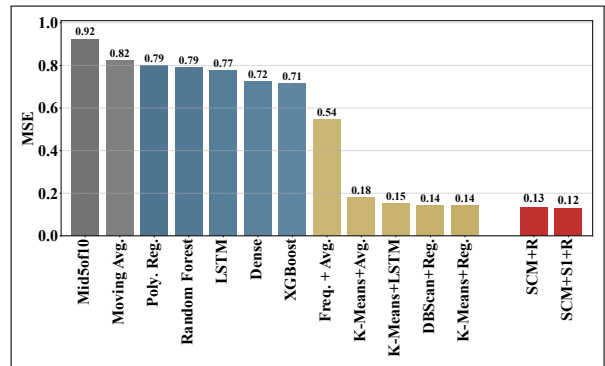


Fig. 3. Baseline estimation over 50 buildings, comparing Averaging (grey), Regression (blue), Clustering (yellow), and Synthetic Control (red) methods.

TABLE II

ACCURACY OF LINEAR SYNTHETIC CONTROL VARIANTS AND THE BEST BENCHMARK OVER 50 RESIDENTIAL BUILDINGS.

| Scenario        | MSE    | Min    | Max    | STD    | Diff   |
|-----------------|--------|--------|--------|--------|--------|
| K-Means + Lasso | 0.1426 | 0.0303 | 0.3610 | 0.0751 |        |
| SCM+R           | 0.1351 | 0.0273 | 0.4077 | 0.0869 | 5.26%  |
| SCM+S1+R        | 0.1274 | 0.0243 | 0.3393 | 0.0815 | 10.66% |
| SCM+U+R         | 0.1275 | 0.0246 | 0.3366 | 0.0818 | 10.60% |

Abbreviations: K-Means + Lasso denotes K-Means clustering with Lasso regression. For the SCM variants, S1 indicates a sum-to-one constraint only, U indicates no constraints, and R denotes regularization.

meteorological variables (temperature, humidity, wind speed, wind direction) and three temporal indicators (weekday, and sine and cosine of hour-of-day).

For the nonlinear SCM, we test multiple MLP configurations, varying model depth, width, learning rate, and batch size. The best-performing model uses two hidden layers of 64 units, followed by a layer of  $J$  units and a scalar output. Training employs a learning rate of  $5 \times 10^{-4}$ , early stopping with a patience of 10, and a maximum of 2000 epochs.

#### C. Setup

We compare benchmark methods and SCM variants across 50 buildings, evaluating classic SCM, augmented SCM, and generalized SCM. Each building is evaluated as the treated unit in turn, with the remaining 299 buildings serving as the control group. Unless otherwise specified, Mean Squared Error (MSE) values are averaged over all treated buildings. To reduce variability from random initialization, neural networks are trained three times per treated building.

Benchmark models are adapted from their original publications, with architectures and hyperparameters tuned to the dataset and baseline estimation task. Additional implementation details can be found in the cited works.

### IV. RESULTS

#### A. Linear Synthetic Control for Baseline Estimation

We begin by benchmarking the linear SCM, serving as the foundation for later extensions. Figure 3 presents the MSE for averaging (grey), regression (blue), and clustering (yellow) benchmarks, compared with our linear SCM (red).

Among the averaging-based baselines, Mid5of10 [17] and Moving Average [17] yield MSE values of 0.9243 and

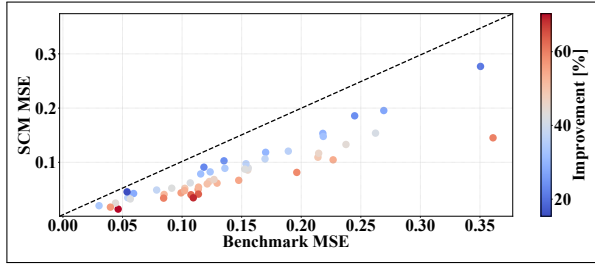


Fig. 4. MSE comparison over 50 buildings between SCM+S+R+Dpast and the best benchmark. Sub-diagonal points show SCM improvements.

0.8250, respectively. Regression-based methods show modest improvements: Polynomial Regression [25] achieves an MSE of 0.7992, followed by Random Forest [11] at 0.7914, LSTM [15] at 0.7756, DNN [17] at 0.7268, and XGBoost [11] at 0.7170. While these reductions are consistent across models, the differences remain within the bounds of statistical variation, with standard deviations ranging from 1.6095 (XGBoost) to 1.8329 (Random Forest).

In our experiments, clustering-based methods that incorporate data from similar buildings consistently achieve lower MSE values than benchmarks based solely on the treated building. K-means and DBSCAN perform comparably, while frequency-based clustering yields slightly higher errors. Among the variants, K-means with simple averaging attains an MSE of 0.1800, pairing K-means with an LSTM reduces it to 0.1534, and combining K-means with Lasso Regression achieves the lowest value of 0.1426.

Table II compares the best-performing benchmark, i.e., K-means with Lasso regression, with three regularized variants of the linear SCM. The SCM+R variant, which applies both sum-to-one and non-negativity constraints along with regularization, achieves an MSE of 0.1351, a 5.26% improvement over the benchmark. The SCM+S1+R variant, retaining only the sum-to-one constraint while relaxing non-negativity, yields the lowest MSE of 0.1274, a 10.66% improvement and reducing the minimum error from 0.0303 to 0.0243 and the maximum from 0.3610 to 0.3393. Finally, the unconstrained SCM+U+R variant, which removes both constraints but retains regularization, attains an MSE of 0.1275, corresponding to a 10.60% improvement.

These results demonstrate that SCM with proper regularization can outperform existing baselines, offering both high accuracy and interpretable weight structures suitable for demand response settlements.

### B. Effect of the Augmented Control Group

We posit that accurate baseline estimation benefits from explicitly modeling both external drivers and temporal dependencies, which are not fully captured by the standard SCM formulation. To demonstrate this, we build on the best-performing linear variant (SCM+S1+R) by expanding the control group in three successive stages: incorporating exogenous features (+exF), adding recent values of the treated unit (+Tpast), and including donor-specific lagged values (+Dpast). Each stage builds on the previous one, resulting

TABLE III  
ACCURACY OF THE GENERALIZED CONTROL GROUP AND THE BEST BENCHMARK OVER 50 RESIDENTIAL BUILDINGS.

| Scenario       | MSE    | Min    | Max    | STD    | Diff   |
|----------------|--------|--------|--------|--------|--------|
| K-Means+Lasso  | 0.1426 | 0.0303 | 0.3610 | 0.0751 |        |
| SCM+S1+R       | 0.1274 | 0.0243 | 0.3393 | 0.0815 | 10.66% |
| SCM+S1+R+exF   | 0.1271 | 0.0245 | 0.3412 | 0.0817 | 10.87% |
| SCM+S1+R+Tpast | 0.0920 | 0.0155 | 0.2912 | 0.0572 | 35.48% |
| SCM+S1+R+Dpast | 0.0821 | 0.0138 | 0.2768 | 0.0519 | 42.43% |

Abbreviations: K-Means + Lasso denotes K-Means clustering with Lasso regression. SCM+S1+R is the regularized SCM with a sum-to-one constraint. +exF adds external features. +Tpast adds external features and recent treated building values. +Dpast adds external features, treated history, and the top correlated past donor value.

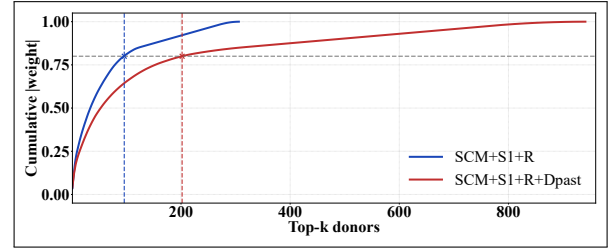


Fig. 5. Cumulative absolute weight share for SCM+S1+R and SCM+S1+R+Dpast. Dashed lines indicate the number of top donors required to account for 80% of the total weight.

in progressively richer representations of the control group. Table III reports the performance of each configuration relative to the strongest benchmark, K-Means + Lasso.

As shown in Table III, the inclusion of external features (+exF) decreases the mean MSE from 0.1274 for SCM+S1+R to 0.1271, corresponding to a marginal gain over SCM+S1+R and a total improvement of 10.87% relative to the benchmark. Augmenting the model with recent values of the treated unit (+Tpast) further reduces the mean MSE to 0.0920, yielding an additional improvement of 27.79% over SCM+S1+R and 35.48% compared to the benchmark. The final extension, incorporating donor-specific lags (+Dpast), attains the lowest mean MSE of 0.0821, corresponding to improvements of 35.56% over SCM+S1+R and 42.43% relative to the benchmark. This configuration also reduces the minimum error from 0.0303 to 0.0138, the maximum error from 0.3610 to 0.2768, and the standard deviation from  $\pm 0.0751$  to  $\pm 0.0519$ , indicating both increased accuracy and reduced variability across buildings.

Figure 4 shows building-level MSE values for the SCM+S1+R+Dpast variant compared to the strongest benchmark. Points below the parity line indicate buildings where the SCM achieves lower errors, with color shading denoting relative improvement. The distribution reveals a consistent advantage for SCM, with lower MSE across all 50 buildings. On average, the mean MSE decreases by 42.43%, with individual gains ranging from 15.37% to 70.31%. Improvements occur across the entire error spectrum, indicating stable performance independent of initial error magnitude.

We further analyze the distribution of control group weights to determine whether the observed gains stem from a few dominant donors or from broader participation. Figure 5

TABLE IV

BASELINE ESTIMATION PERFORMANCE OF LINEAR AND NONLINEAR SCM  
WITH BASIC AND AUGMENTED CONTROL GROUPS

| Scenario       | MSE    | Min    | Max    | STD    | Diff   |
|----------------|--------|--------|--------|--------|--------|
| K-Means+Lasso  | 0.1426 | 0.0303 | 0.3610 | 0.0751 |        |
| SCM+S1+R       | 0.1274 | 0.0243 | 0.3393 | 0.0815 | 10.66% |
| NN SCM         | 0.1260 | 0.0256 | 0.3636 | 0.0852 | 11.64% |
| SCM+S1+R+Dpast | 0.0821 | 0.0138 | 0.2768 | 0.0519 | 42.43% |
| NN SCM+Dpast   | 0.0817 | 0.0127 | 0.2796 | 0.0543 | 42.70% |

Abbreviations: K-Means + Lasso denotes K-Means clustering with Lasso regression. SCM+S1+R is the regularized SCM with a sum-to-one constraint. NN SCM is the generalized neural network extension. +Dpast adds external features, treated history, and the top correlated past donor value.

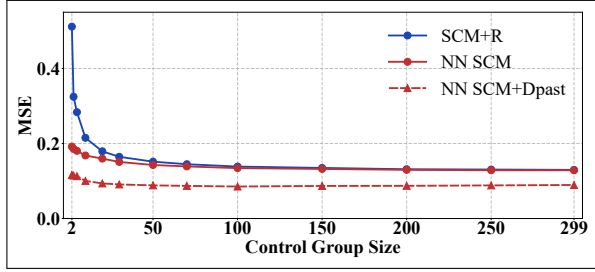


Fig. 6. Comparison of MSE between linear and nonlinear SCM variants for different control group sizes.

compares cumulative weight distributions for SCM+S1+R and SCM+S1+R+Dpast, ranking donors by absolute weight. In SCM+S1+R, 80% of the total absolute weight is concentrated among the top 96 donors, with the largest individual weight of 0.1329. In contrast, SCM+S1+R+Dpast requires 202 donors to reach the same 80% threshold, indicating a more even distribution, though the maximum weight increases slightly to 0.1563. The observations suggest that the improved accuracy may, in part, arise from a broader and more balanced donor contribution, reducing dependence on dominant units and improving stability.

### C. Effect of the Generalized Nonlinear Weighting

To increase SCM flexibility, the linear weighting is replaced with a nonlinear mapping that captures complex dependencies between the treated unit and its donors. The DNN-based approach is evaluated for both the standard and augmented control groups.

Table IV shows that both nonlinear and linear variants achieve nearly identical accuracy. For the standard control group, the mean MSE decreases from 0.1274 (linear) to 0.1260 (nonlinear), a relative improvement of 1.10%. For the generalized control group, it decreases from 0.0821 to 0.0817, an improvement of 0.49%. Both variants outperform the benchmark substantially, improving accuracy by 42.43% (linear) and 42.71% (nonlinear).

To analyze robustness, we further evaluate both weighting schemes across progressively smaller control groups, as summarized in Figure 6.

For both linear and nonlinear SCM variants, expanding the donor pool consistently lowers the MSE. As the donor pool shrinks, performance gaps widen. In the extreme case of two donors, the error increases sharply to 0.5116 for SCM+R

but remains considerably lower at 0.1918 for NN SCM and 0.1163 for NN SCM+Dpast.

These results suggest that while both methods perform comparably with large donor pools, the nonlinear SCM maintains lower errors and reduced variability when the control group is small.

## V. DISCUSSION

We observed that across all experiments, the generalized SCM demonstrates improvements in baseline estimation accuracy and robustness.

In Section IV-A, the standard linear SCM exceeds the benchmark accuracy. Unlike clustering-based methods that preselect donors by similarity, the standard SCM optimizes weights over the entire donor pool. This allows it to assign weights to donors that may be individually dissimilar to the treated unit, whose combination nonetheless closely approximates the treated unit's behavior. Relaxing the non-negativity constraint allows the model to exploit both positive and negative correlations in demand, increasing flexibility in reconstructing counterfactual load profiles. This likely contributes to the strong performance observed even in heterogeneous control groups.

Section IV-B extends the control group with external features, recent treated unit history, and lagged donor values. External features add limited benefit, whereas lagged treated values provide an autoregressive signal that compensates for limited or imperfect donor matches. Further discussion is supplied in the Appendix. Lagged donor values align similar patterns occurring at different times, improving matching under temporal misalignment.

In Section IV-C, the nonlinear weighting matches the accuracy of the linear SCM for large donor pools, indicating linear sufficiency in well-represented settings. When the donor pool is small or less representative, however, the nonlinear variant achieves lower errors and reduced variance. The nonlinear weighting thus increases robustness in data-constrained scenarios.

### A. Limitations

Here, we discuss limitations related to data scope, feature representation, and benchmarking, while highlighting directions for methodological improvement. The evaluation relies on a single dataset that, although representative of realistic operating conditions, may limit generalizability to other regulatory contexts or flexibility mechanisms. Integrating weather and calendar features yielded only marginal gains, likely due to the low temporal resolution of the weather data (4-hour intervals); higher-resolution or localized data may produce stronger effects. Benchmarking remains challenging given the heterogeneity of existing methods and their differing assumptions, such as prior knowledge of activation periods. To ensure comparability, we selected and adapted relevant baselines from the literature within our generalized framework. Finally, computational constraints limited hyperparameter optimization to a feasible subset of configurations, suggesting potential gains from further tuning. Despite these

limitations, the proposed SCM consistently outperformed all benchmarks across evaluation scenarios.

## VI. CONCLUSIONS AND FUTURE WORK

We proposed a Generalized Synthetic Control framework for baseline estimation in residential demand response, extending the classical SCM with an augmented control representation and a nonlinear weighting function. On the Ausgrid dataset, the linear SCM outperformed the strongest benchmarks by more than 10%, while the generalized variant improved accuracy by up to 42%. Linear weighting remained highly effective with large donor pools, whereas the nonlinear model increased robustness and accuracy for smaller control groups, supporting effective use of demand-side flexibility for congestion management.

Future work should broaden evaluation across diverse grid topologies, market structures, and demand-side behaviors to assess generalizability and robustness. Methodologically, generating synthetic donors in data-sparse or low-diversity settings, e.g., via variational autoencoders, generative adversarial networks, or diffusion models—could enrich the control group and improve counterfactual accuracy and stability. Finally, incorporating uncertainty quantification to produce confidence intervals for counterfactual trajectories would enable risk-aware flexibility assessments and more transparent settlements.

## APPENDIX

### A. Linear Program for Energy Management System

This section outlines the linear program used to optimize BESS operations under dynamic electricity prices from the Australian Energy Market Operator [26]. The Energy Management Systems (EMS) minimizes total electricity costs, accounting for both consumption and flexibility revenues.

*Objective Function:*

$$\min \sum_{t \in T} (p_t \cdot x_t - f_t \cdot \delta_t^+) \quad (9)$$

with dynamic electricity price  $p_t$ , net grid consumption  $x_t$ , flexibility rebate  $f_t$ , and rewarded reduction  $\delta_t^+$ .

*Constraints:*

Net grid consumption  $x_t$  includes building load  $l_t$ , PV generation  $g_t$ , and BESS operation  $s_t$ .

$$x_t = l_t - g_t + s_t, \quad \forall t \in T \quad (10)$$

The baseline  $b_t$  is the average consumption during the same hour of the  $n$  most recent congestion-free days:

$$b_t = \frac{1}{n} \sum_{k \in S_t} x_{t-24k}, \quad \forall t \in T, \quad \forall k \in \{1, \dots, n\} \quad (11)$$

$$S_t = \left\{ k \in \{1, 2, \dots, n\} \mid \sum_{h=-12}^{12} C_{t-24k+h} = 0 \right\} \quad (12)$$

Flexibility  $\delta_t^+$  is the positive deviation from the baseline.

$$\delta_t^+ = \max(0, b_t - x_t), \quad \forall t \in T \quad (13)$$

The State of Charge (SOC) is updated with dis-/ charging actions  $s_t$ , considering efficiency  $\eta_s$ .

$$\text{SOC}_t = \text{SOC}_{t-1} + \eta_s \cdot s_t, \quad \forall t \in T \setminus \{0\}. \quad (14)$$

Domain constraints ensure feasible operation:

$$s_t \in [-s^{\max}, s^{\max}], \quad \text{soc}_t \in [\text{soc}^{\min}, \text{soc}^{\max}] \quad (15)$$

### B. Theoretical Justification of Augmented Synthetic Control Estimator

In this section we provide a theoretical justification for the Augmented Synthetic Control (ASC) estimator, specifically, for including the lagged values of the treated and donor units in the augmented matrix. Let  $y_{it}$  be the outcome for unit  $i$  at time  $t$ , and let

$$SC_t = \sum_{j=2}^{J+1} w_j^* y_{jt}$$

be the value of the standard synthetic control at time  $t$ . Consider the following ASC estimator for the counterfactual:

$$\hat{y}_{1,T_0+1} = SC_{T_0+1} + \beta_1 y_{1,T_0} + \beta_2 SC_{T_0} \quad (16)$$

While this formulation is slightly less general than (6)–(8), it nonetheless offers adequate generality for theoretical insights within our scope. Its causal interpretation is limited to the  $T_0+1$  period, as any further extension would rely on post-treatment information. We next examine two theoretically motivated restrictions of this framework: (i) an autoregressive correction and (ii) an error-correction mechanism.

1) *The Autoregressive (AR) Correction:* The first approach assumes that the fitting error is best predicted by the treated unit's own past momentum and is achieved by setting  $\beta_2 = 0$  in (16):

$$\hat{y}_{1,T_0+1} = SC_{T_0+1} + \beta_1 y_{1,T_0} \quad (17)$$

where the coefficient  $\beta_1$  is estimated from the pre-treatment data by regressing the fitting errors,

$$e_{1t} = y_{1t} - SC_t,$$

on the lagged outcome of the treated unit,  $y_{1,t-1}$ .

**Proposition 1.** Let the AR-Augmented Synthetic Control estimator be given by (17), with  $\beta_1$  estimated on the pre-treatment window  $T_0$  by OLS, regressing the fitting error  $e_{1t} = y_{1t} - SC_t$  on the lagged outcome  $y_{1,t-1}$ . If  $\beta_1 \neq 0$ , then the estimator has strictly lower mean squared error (MSE) at  $T_0+1$  than the standard Synthetic Control estimator.

**Proof sketch.** Define the one-step SC error  $e_{SC} := y_{1,T_0+1} - SC_{T_0+1}$ . Pre-treatment OLS finds  $\beta_1$  that best linearly predicts  $e_{1t}$  from  $y_{1,t-1}$ . Assuming this one-step forecasting relation is stable at  $T_0+1$ , we use the same  $\beta_1$  to form the correction  $\beta_1 y_{1,T_0}$ . The AR-ASC error is

$$e_{ASC} = e_{SC} - \beta_1 y_{1,T_0}.$$

Because  $\beta_1 \neq 0$ , the correction removes a nonzero predictable component of  $e_{SC}$ . In linear prediction terms,

subtracting the best linear forecast strictly reduces variance/MSE:

$$\text{Var}(e_{ASC}) = \text{Var}(e_{SC} - \beta_1 y_{1,T_0}) < \text{Var}(e_{SC}).$$

Hence  $\text{MSE}(\text{AR-ASC}) < \text{MSE}(\text{SC})$ .

2) *The Error-Correction Mechanism (ECM)*: The second approach assumes that the *gap* between the treated unit and its synthetic counterpart persists over time and seeks to correct it. This corresponds to the classic error-correction mechanism in time-series econometrics. This interpretation is achieved by setting  $\beta_1 = -\beta_2 = \rho$  in (16):

$$\hat{y}_{1,T_0+1} = SC_{T_0+1} + \rho(y_{1,T_0} - SC_{T_0})$$

where  $\rho$  denotes the rate of error persistence, which can be estimated from the pre-treatment data.

**Proposition 2.** *Assume the counterfactual outcome for the treated unit is generated by a process with an idiosyncratic, stationary AR(1) shock,  $\epsilon_{1t} = \rho\epsilon_{1,t-1} + u_{1t}$ . The ECM-Augmented Synthetic Control estimator has a strictly lower MSE than the standard Synthetic Control estimator.*

**Proof Sketch.** The estimation error of the standard SC estimator,  $e_{SC}$ , is the difference between the idiosyncratic shocks of the treated unit and its synthetic counterpart:

$$e_{SC} = \epsilon_{1,T_0+1} - \sum w_j^* \epsilon_{j,T_0+1}.$$

Under the AR(1) assumption for the treated unit's shock,  $\epsilon_{1,T_0+1} = \rho\epsilon_{1,T_0} + u_{1,T_0+1}$ , this error contains a predictable component,  $\rho\epsilon_{1,T_0}$ . The standard estimator is therefore sub-optimal, as it fails to account for this persistence.

The ECM estimator introduces a correction term,  $\rho(y_{1,T_0} - SC_{T_0})$ . Similarly, the error of the ECM estimator,  $e_{ASC}$ , is derived from first principles as  $e_{ASC} = e_{SC} - \rho(y_{1,T_0} - SC_{T_0})$ .

The observable pre-treatment gap  $(y_{1,T_0} - SC_{T_0})$  serves as a consistent empirical proxy for the unobserved shock gap at time  $T_0$ , which is dominated by  $\epsilon_{1,T_0}$ . Hence, the correction term acts as an empirical analog of the predictable error component,  $\rho\epsilon_{1,T_0}$ . Subtracting this correction from the total error effectively cancels the predictable component. The resulting estimation error of the augmented estimator,  $e_{ASC}$ , is thus purged of the correlated component attributable to the treated unit. Consequently, the variance of  $e_{ASC}$  is strictly lower than that of  $e_{SC}$ , as the dominant predictable portion of the idiosyncratic shock has been removed. This reduction in error variance directly implies a lower MSE for the ECM estimator.

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